SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I Bharath P(192511282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Bharath

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Industry Problem: Diabetes Diagnosis and Personalised Treatment Recommendation

Introduction

A healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI and family history to develop an AI-based system for early diabetes diagnosis. The system should classify whether a patient is diabetic or non-diabetic and should also recommend suitable treatment actions such as Diet, Exercise, Medication and Monitor. The first part can be solved using supervised machine learning techniques such as Decision Tree and Logistic Regression, while the treatment recommendation can be formulated as a Markov Decision Process and solved using Q-Learning. The complete solution should focus particularly on reducing false-negative diabetes cases, maintaining model interpretability and ensuring patient safety.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from previously observed examples and uses those patterns to predict unseen cases. For this medical dataset, the problem can be formulated as a binary classification problem.

Target Variable:

Diabetes = 1 → Diabetic

Diabetes = 0 → Non-diabetic

Important Features:

Feature	Justification
Glucose Level	High glucose level is strongly associated with diabetes and is an important predictor.
BMI	Higher BMI can be associated with increased diabetes risk.
Age	Diabetes risk can vary with age and therefore age is useful for classification.
Blood Pressure	Provides additional information about the patient's health condition.
Cholesterol	Provides information about metabolic and cardiovascular health.
Family History	A family history of diabetes can indicate increased genetic risk.

The three most important initial features would generally be glucose level, BMI and age, although the final importance should be determined from the trained model.

Inductive Learning Process:

Collect patient records.

Clean and preprocess the data.

Separate input features and the diabetes target.

Divide the data into training and testing sets.

Train a classification model.

Validate the model using appropriate performance measures.

Use the trained model to predict diabetes for new patients.

Thus, the model learns a general relationship between patient characteristics and diabetes status from the available labelled examples.

(b) Handling Class Imbalance

The dataset may contain significantly more non-diabetic patients than diabetic patients. This creates a class imbalance problem. A model trained directly on such data may become biased toward the majority class and predict most patients as non-diabetic.

Suitable Strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) is a suitable approach. SMOTE creates synthetic examples of the minority diabetic class based on existing minority-class observations.

Procedure:

Identify the minority diabetic samples.

Select neighbouring diabetic samples.

Generate synthetic samples between existing minority observations.

Increase the representation of the diabetic class.

Train the model using the balanced training data.

SMOTE is useful because the main clinical concern is detecting diabetic patients correctly. Missing a diabetic patient is a false negative, which can be more serious than incorrectly flagging a non-diabetic patient. SMOTE must be applied only to the training data, after the train-test split, to avoid data leakage.

Class weights can also be used as an alternative. Under-sampling is less desirable if it discards useful medical information from the majority class.

(c) 70:30 Train-Test Split and Preprocessing

The dataset can be divided as follows:

70% → Training data

30% → Testing data

The 70:30 split provides enough data for the model to learn patterns while keeping a sufficiently large independent test set for evaluating performance. For a medical application, the test set should not be used during training because it must represent unseen patients. A stratified split should be used so that the diabetic/non-diabetic ratio is approximately preserved in both training and testing sets.

Preprocessing Steps:

1. Missing Value Handling: Missing numerical values can be handled using median imputation, while categorical missing values can be replaced using the most frequent category.

2. Normalisation: Numerical features such as age, blood pressure, cholesterol, glucose and BMI can be normalised when using models sensitive to feature scale.

$z = (x - \mu) / \sigma$

where x is the original value, μ is the mean and σ is the standard deviation.

3. Encoding: Family history can be encoded as No = 0 and Yes = 1. The target can similarly be represented as Non-diabetic = 0 and Diabetic = 1.

Impact: Preprocessing improves data quality, makes the input suitable for machine learning and prevents numerical differences between features from unnecessarily affecting scale-sensitive algorithms.

TASK 2: DECISION TREE FOR DIAGNOSIS

(a) Decision Tree Classifier

A Decision Tree classifies patients by repeatedly dividing the dataset according to feature values. For the diabetes problem, the tree can use glucose, BMI, age, blood pressure, cholesterol and family history.

Illustrative first three levels of the tree:

```
                    Glucose
               /           \
          ≤ 125              > 125
          /                     \
         BMI                  Glucose
     /      \               /      \
  ≤ 30      >30          ≤ 180    >180
    /          \           /         \
Non-diabetic     Age       Monitor      Diabetic
               /   \
            ≤ 50    >50
```

| |

Non-diabetic Diabetic

The exact tree depends on the actual patient dataset.

Root Node Selection:

The root node should be selected using the feature that produces the largest reduction in impurity. One method is Information Gain.

$IG(S,A) = H(S) - \Sigma(|S_v|/|S|)\ H(S_v)$

$H(S) = -\Sigma\ p_i \log_2(p_i)$

Another method is the Gini Index:

$Gini(S) = 1 - \Sigma\ p_i^2$

The feature with the highest Information Gain or lowest resulting Gini impurity is selected as the root.

Feature	**Illustrative Information Gain**
Glucose	0.31
BMI	0.18
Age	0.12
Blood Pressure	0.08
Cholesterol	0.06
Family History	0.10

Since glucose has the highest illustrative Information Gain, IG(Glucose) = 0.31, glucose is selected as the root node. These values are illustrative because the assessment file does not provide a numerical patient dataset.

(b) Model Evaluation

The Decision Tree should be evaluated using Accuracy, Precision, Recall and F1-Score.

$Accuracy = (TP + TN) / (TP + TN + FP + FN)$

$Precision = TP / (TP + FP)$

$Recall = TP / (TP + FN)$

$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$

Illustrative Confusion Matrix:

	Actual Diabetic	**Actual Non-diabetic**
Predicted Diabetic	TP = 25	FP = 5
Predicted Non-diabetic	FN = 3	TN = 67

Total = 25 + 5 + 3 + 67 = 100

Accuracy = (25 + 67) / 100 = 0.92 = 92%

Precision = 25 / (25 + 5) = 0.8333 = 83.33%

Recall = 25 / (25 + 3) = 0.8929 = 89.29%

F1 = 2 × (0.8333 × 0.8929) / (0.8333 + 0.8929) ≈ 0.862 = 86.2%

False Negatives = 3. These are patients who actually have diabetes but are incorrectly classified as non-diabetic.

False negatives can be reduced by using class weights, applying SMOTE to the training set, adjusting the classification threshold, optimising the model for recall, collecting more representative diabetic samples and using a second clinical test for high-risk cases. Therefore, the system should not rely only on accuracy; recall and false-negative rate should receive strong consideration.

(c) Pre-Pruning vs Post-Pruning

A Decision Tree can become too complex and overfit the training data.

Pre-pruning limits tree growth during training using maximum tree depth, minimum samples required for splitting and minimum samples per leaf. It provides faster training and a smaller tree, but important patterns may be removed too early.

Post-pruning first builds a larger tree and then removes branches that provide insufficient predictive benefit. One common method is cost-complexity pruning.

$R\alpha(T) = R(T) + \alpha|T|$

where $R(T)$ is tree error, $|T|$ is the number of terminal nodes and α is the complexity parameter.

Model	Illustrative Accuracy	Complexity
Fully grown Tree	94% training / 84% testing	High
Pre-pruned Tree	90%	Medium
Post-pruned Tree	92%	Low-Medium

The fully grown tree may achieve high training accuracy but lower testing accuracy because of overfitting. A post-pruned Decision Tree is preferable for deployment when it maintains good predictive performance while reducing unnecessary complexity and remains easier for healthcare professionals to interpret.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Logistic Regression

Logistic Regression can be applied to the same diabetes dataset as a statistical learning model. The probability of diabetes is:

$P(Y = 1) = 1 / (1 + e^{(-z)})$

where $z = \beta_0 + \beta_1X_1 + \beta_2X_2 + ... + \beta_nX_n$.

For this problem, $z = \beta_0 + \beta_1(\text{Glucose}) + \beta_2(\text{BMI}) + \beta_3(\text{Age}) + \beta_4(\text{BP}) + \beta_5(\text{Cholesterol}) + \beta_6(\text{FamilyHistory})$.

For example, if P(Diabetes) = 0.82 and the selected threshold is 0.5, then 0.82 > 0.5 and the patient is classified as diabetic.

Model	Illustrative Accuracy	Illustrative AUC-ROC
Decision Tree	92%	0.90
Logistic Regression	94%	0.93

The values above are illustrative because the assessment file does not provide a numerical patient dataset from which actual model results can be calculated.

AUC-ROC represents the model's ability to distinguish between diabetic and non-diabetic patients. An AUC closer to 1 indicates excellent discrimination, while AUC = 0.5 indicates performance close to random classification.

Logistic Regression may be preferable when probability estimates are important, interpretability is required, the effect of individual features needs to be studied, and a simpler model is preferred for clinical risk scoring. Decision Trees are preferable when non-linear relationships and easy rule-based interpretation are important.

(b) Feature Importance Analysis

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The models agree that glucose and BMI are among the strongest predictors. The third predictor may differ because the two models measure relationships differently. A Decision Tree measures reduction in impurity produced by a feature during splitting, whereas Logistic Regression estimates the relationship between predictors and the log-odds of the outcome. Therefore, differences in the top-three rankings are possible.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) Modelling Treatment Recommendation as an MDP

The treatment recommendation problem can be represented using a Markov Decision Process (MDP). An MDP consists of (S, A, R, P, γ), where S represents states, A represents actions, R represents rewards, P represents transition probabilities and γ represents the discount factor.

1. States:

State	**Description**
S1	Low-risk / controlled
S2	Moderate-risk
S3	High-risk
S4	Poorly controlled
S5	Improved / stable

2. Actions: A = {Diet, Exercise, Medication, Monitor}. These are the actions specified in the assessment.

3. Reward Function: The reward should represent improvement in patient health.

Outcome	**Reward**
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	−10

4. Transition Dynamics: Transition dynamics describe how the patient moves from one state to another after an action. $P(S_{t+1} \mid S_t, A_t)$ represents the probability of reaching the next state given the current state and selected action.

Example: Moderate Risk + Diet → Improved / Stable. High Risk + Monitor → High Risk. In a real system, transition probabilities must be estimated from reliable longitudinal clinical data.

(b) Q-Learning

Q-Learning is a model-free reinforcement learning algorithm that learns the expected future reward for each state-action pair.

$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

where Q(s,a) is the current Q-value, α is the learning rate, r is the reward, γ is the discount factor, s′ is the next state and max Q(s′,a′) is the best future Q-value.

Assume $\alpha = 0.5$, $\gamma = 0.9$ and initially all Q-values are 0.

Five illustrative patient episodes:

Episode	**Initial State**	**Action**	**Next State**	**Reward**
1	Moderate Risk	Diet	Improved	+10

2	Moderate Risk	Exercise	Improved	+8
3	High Risk	Medication	Moderate Risk	+6
4	Moderate Risk	Monitor	Moderate Risk	0
5	High Risk	Exercise	High Risk	−2

These episodes are an illustrative demonstration of the required Q-learning procedure and are not clinical recommendations.

Q-Table Update – Iteration 1

1. Moderate Risk + Diet: Qnew = 0 + 0.5[10 + 0.9(0) − 0] = 5. Therefore Q(Moderate, Diet) = 5.

2. Moderate Risk + Exercise: Qnew = 0 + 0.5[8 + 0.9(0) − 0] = 4. Therefore Q(Moderate, Exercise) = 4.

3. High Risk + Medication: Qnew = 0 + 0.5[6 + 0.9(0) − 0] = 3. Therefore Q(High, Medication) = 3.

State	**Diet**	**Exercise**	**Medication**	**Monitor**
Moderate Risk	5.0	4.0	0	0
High Risk	0	0	3.0	0

Q-Table Update – Iteration 2

1. Moderate Risk + Exercise: Current Q = 4, r = 8, and max next-state Q = 5.

Qnew = 4 + 0.5[8 + 0.9(5) − 4] = 4 + 0.5(8.5) = 8.25.

2. High Risk + Medication: Current Q = 3, r = 6, and max next-state Q = 5.

Qnew = 3 + 0.5[6 + 0.9(5) − 3] = 3 + 0.5(7.5) = 6.75.

3. Moderate Risk + Diet: Current Q = 5, r = 10, and max next-state Q = 0.

Qnew = 5 + 0.5[10 + 0.9(0) − 5] = 7.50.

State	**Diet**	**Exercise**	**Medication**	**Monitor**
Moderate Risk	7.50	8.25	0	0
High Risk	0	0	6.75	0

Convergence Criterion:

The Q-learning process can be considered approximately converged when the maximum change in Q-values between successive iterations becomes smaller than a predefined threshold, for example max |Qnew − Qold| < 0.01 for a sufficiently large number of consecutive updates. Another practical criterion is that the learned policy remains unchanged over several episodes.

(c) Final Learned Policy

The policy is obtained by selecting the action with the maximum Q-value for each state:

$\pi(s) = \arg\max_a Q(s,a)$

For Moderate Risk: Q(Diet) = 7.50 and Q(Exercise) = 8.25. Therefore π(Moderate Risk) = Exercise.

For High Risk: Q(Medication) = 6.75. Therefore π(High Risk) = Medication.

Patient State	**Illustrative Learned Action**
Moderate Risk	Exercise
High Risk	Medication

This is only an example of how a policy is extracted from a Q-table. It must not be interpreted as a real medical treatment recommendation. In an actual healthcare system, treatment decisions must be determined and supervised by qualified healthcare professionals.

Ethical Considerations:

Patient Safety: Incorrect recommendations can cause harm. The AI system should not independently make unsafe clinical decisions.

Bias: If the training data is biased or does not adequately represent different patient groups, the model may produce unfair recommendations.

Explainability: Healthcare professionals should understand why the system selected a particular recommendation.

Privacy: Patient health records contain sensitive information, so security, access control and privacy protection are required.

Human Oversight: The RL agent should operate as a decision-support system, not as an uncontrolled autonomous treatment system.

Validation: The model must be tested extensively before deployment for safety, reliability, bias and performance.

Overall Conclusion

The proposed AI system combines supervised machine learning and reinforcement learning to support diabetes diagnosis and treatment decision-making. Inductive learning can use patient features such as glucose level, BMI, age, blood pressure, cholesterol and family history to predict diabetes. A 70:30 stratified train-test split, appropriate preprocessing and class-balancing techniques such as SMOTE can improve model development. A Decision Tree provides an interpretable classification model, while Logistic Regression provides a statistical alternative with useful probability and risk estimates.

For medical diagnosis, accuracy alone is insufficient because false-negative diabetes cases can be clinically important. Therefore, recall, precision, F1-score and AUC-ROC should all be considered. Post-pruning can reduce Decision Tree complexity and improve generalisation.

For treatment recommendation, the problem can be represented as an MDP containing patient states, treatment actions, rewards and transition dynamics. Q-Learning can then learn the expected value of different actions through repeated episodes. The final policy is obtained by selecting the action with the highest Q-value.

However, an AI system for healthcare must not replace qualified medical professionals. Patient safety, fairness, privacy, explainability and human supervision are essential before any such system is deployed. The final system should therefore be designed as a validated clinical decision-support tool, with medical professionals retaining responsibility for patient care.

Note on Numerical Results

The assessment document specifies the required tasks and methods but does not include an actual numerical patient dataset. Therefore, the Information Gain values, confusion-matrix values, model comparison values and Q-learning episodes used above are illustrative examples for demonstrating the required calculations. Actual experimental results should be generated from the supplied patient dataset if one is provided.